

Intel Pentium Microprocessor Architecture

Improvements in Pentium (Compared to 80486)

Key Points

- Split cache: **8 KB Instruction Cache + 8 KB Data Cache**
- **64-bit data bus** (bigger transfer per cycle)
- **Faster FPU** (better floating-point performance)
- **Dual integer pipelines** (superscalar: up to 2 integer instructions/cycle)
- **Branch prediction** (fewer pipeline stalls)
- **MMX instructions** (multimedia/packed data)

Split Cache Structure

What it is: Pentium uses two separate caches:

- **Instruction Cache (8 KB)** for instructions
- **Data Cache (8 KB)** for data

Why it matters: Instruction fetch and data access can happen in parallel, reducing delays.

64-bit Data Bus

What it is: Bus width increases from **32-bit** (80486) to **64-bit** (Pentium). **Why it matters:** More bytes move per memory cycle $\Rightarrow$ faster overall performance.

Faster Numeric Coprocessor (FPU)

What it is: The floating-point unit is significantly faster than 80486. **Why it matters:** Improves scientific computing, graphics, engineering calculations.

Dual Integer Processor (Superscalar)

What it is: Two integer pipelines enable **instruction-level parallelism**. **Why it matters:** Up to **two suitable** integer instructions can execute in one clock cycle.

Branch Prediction Logic

What it is: CPU predicts branch direction/target and fetches early. **Why it matters:** Reduces pipeline stalls caused by conditional jumps and loops.

MMX Instructions

What it is: MMX adds packed-data instructions for multimedia tasks. **Why it matters:** Faster image/audio/video style operations.

Exam Line

Pentium boosts speed using **split I/D cache (8 KB + 8 KB)**, **64-bit data bus**, **faster FPU**, **superscalar dual integer pipelines**, **branch prediction**, and **MMX**.

Pentium Internal Architecture (Conceptual View)

Big Idea

Pentium achieves high performance by combining **parallel execution**, **fast caching**, and **smarter control logic** (especially for branches and interrupts).

Dual Processing Logic (U-pipe and V-pipe)

What it is: Two integer pipelines:

- **U-pipe**
- **V-pipe**

Why it matters: Executes independent integer instructions in parallel.

Instruction Fetch and Decode

What it is:

- Prefetch buffers fetch instructions early
- Instruction cache provides fast instruction delivery
- Decode unit dispatches instructions to pipelines

Why it matters: Keeps pipelines busy, reduces waiting.

Branch Hardware (BTB)

What it is: **Branch Target Buffer (BTB)** stores recent branch targets. **Why it matters:** Helps branch prediction fetch correct target faster.

TLB and APIC (Support Blocks)

TLB: speeds up virtual → physical address translation. **APIC:** improves interrupt handling in advanced systems. **Why it matters:** Less overhead for memory management and interrupts.

Exam Line

Pentium's block design includes **U/V pipelines**, **split caches**, **BTB**, **TLB**, and **APIC** to increase throughput and reduce stalls.

Pentium Pro Enhancements

Key Points

- **14-stage superpipelined** architecture
- Can schedule up to **5 instructions**
- **256 KB / 512 KB L2 cache**
- **ECC**: corrects 1-bit, detects 2-bit errors
- **36-bit addressing** $\Rightarrow$ **64 GB** physical memory

Deep Pipeline (Superpipelining)

What it is: 14 pipeline stages. **Why it matters:** Higher clock speed and more overlapping work.

Instruction Scheduling

What it is: Up to 5 instructions can be scheduled. **Why it matters:** Better use of execution units (less idle time).

L2 Cache and ECC

What it is:

- L2 cache reduces main memory access
- ECC improves memory reliability

Why it matters: Important for server/workstation stability and performance.

36-bit Addressing

What it is: 36-bit physical addressing. **Why it matters:** Supports **64 GB** memory (much larger than 4 GB).

Exam Line

Pentium Pro improves Pentium via **14-stage pipeline**, **multi-instruction scheduling**, **large L2 cache**, **ECC**, and **36-bit (64 GB) addressing**.

Memory System Organization

Memory Space

- Total addressable memory: **4 GB**
- Address range: 00000000H to FFFFFFFFH

Banked Memory (8 Banks)

What it is: Memory is organized into **8 banks**. Each bank stores:

- **1 byte** of data
- **1 parity bit**

Why it matters: Matches the **64-bit bus** (8 bytes transferred together):

$$64 \text{ bits} = 8 \text{ bytes} \Rightarrow 8 \text{ banks} \times 1 \text{ byte}$$

Supported Access Sizes

What it supports:

- Byte (8-bit), Word (16-bit), Doubleword (32-bit), Quadword (64-bit)

Why it matters: Flexible data sizes with efficient transfer on a wide bus.

Exam Line

Pentium memory uses **4 GB address space** with **8 bank organization** (byte-wide + parity), supporting efficient **64-bit** transfers.

Memory and I/O System

64-bit Data Bus Advantage

What it is: 64-bit data bus. **Why it matters:** A 64-bit double-precision value can be fetched in one read cycle.

Address Parity Checking

What it is: Parity check on address lines **A31–A5**. **Why it matters:** Detects address-bus parity errors (system/OS decides what to do).

I/O Addressing and Protection

What it is:

- I/O port numbers appear on **A15–A3**
- I/O space size:

$$2^{13} = 8192 \approx 8 \text{ KB}$$

- I/O privilege stored in **TSS**
- Blocked I/O access $\Rightarrow$ **Type 13 interrupt**

Why it matters: Prevents unauthorized I/O operations and improves OS-level protection.

Exam Line

Pentium provides **8 KB I/O space** (A15–A3) with **TSS-based I/O privilege**; violations generate **Type 13 interrupt**. Address parity is checked on **A31–A5**.

Branch Prediction Logic

Quick Summary

- Predict branch target early
- Prefetch target instructions into cache
- Wrong prediction adds extra cycles (about 3)
- Uses **Branch Target Buffer (BTB)**

How It Works

What it is: When a branch instruction appears, the CPU starts fetching from the predicted target address. **Why it matters:** If prediction is correct, instructions are ready and execution is faster.

Branch Target Buffer (BTB)

What it is: BTB stores recent branch targets for dynamic prediction. **Why it matters:** Speeds up fetching the next correct instruction stream.

Exam Line

Branch prediction + BTB reduces branch delay by prefetching target instructions; incorrect prediction costs extra cycles (about 3).

Cache Structure and Superscalar Execution

Quick Summary

- Two caches: **8 KB Instruction + 8 KB Data**
- Three execution units: **FPU + U-pipe + V-pipe**
- Good instruction scheduling can improve speed (up to **40%**)

Split Cache

What it is: separate I-cache and D-cache (8 KB each). **Why it matters:** Parallel instruction/data access improves throughput.

Superscalar Execution Units

What it is:

- 1 Floating Point execution unit
- 2 Integer pipelines (U and V)

Why it matters: Multiple instructions can be executed per clock cycle.

Software Scheduling Note

What it means: Try to separate dependent instructions by placing independent ones in between.

Why it matters: Keeps both integer pipelines busy and improves speed.

Exam Line

Pentium is superscalar: **U/V integer pipelines + FPU** with **split caches**, enabling multiple instructions per cycle and higher performance.

Paging Unit (4 KB and 4 MB Pages)

Core Concept

Paging converts **linear (virtual) addresses** into **physical addresses** using page directory/table structures. Pentium supports both **4 KB** and **4 MB** pages.

4 KB Paging

Address split:

- Page Directory index: 10 bits
- Page Table index: 10 bits
- Offset: 12 bits

Why it matters: Fine-grained mapping; requires both directory and table lookup.

4 MB Paging (Extension)

Address split:

- Page Directory index: 10 bits
- Offset: 22 bits

Key change: No page table needed. **Why it matters:** Faster translation and less overhead for large memory blocks.

PSE Bit

What it is: Page Size Extension (PSE) bit selects 4 KB vs 4 MB pages. **Why it matters:** Allows switching between small and large page mappings.

Example Translation

- Linear address: 00200001H
- Physical address: 01000001H

Exam Line

Pentium supports **4 KB** paging (dir+table+offset) and **4 MB** paging (dir+offset only), controlled using the **PSE** bit.

Quick Revision Table

Topic	Key Feature	Why It Matters
Cache	Split 8 KB I-cache + 8 KB D-cache	Parallel fetch + access; fewer stalls
Data Bus	64-bit data bus	Higher data transfer per cycle
Execution	U/V pipelines (superscalar)	Up to 2 integer instructions per cycle
Branching	Branch prediction + BTB	Reduces branch penalty/stalls
Pentium Pro	14-stage, L2, ECC, 36-bit addr	Speed + reliability + 64 GB memory
Paging	4 KB + 4 MB (PSE)	Flexible mapping; faster large-page translation

Final 3-Line Summary

- Pentium increases performance using **superscalar execution**, **split caches**, and **branch prediction**.
- Memory performance improves via a **64-bit data bus** and **8-bank memory organization**.
- Pentium Pro adds **deep pipelining**, **large L2 cache**, **ECC**, and **36-bit addressing** for large reliable systems.